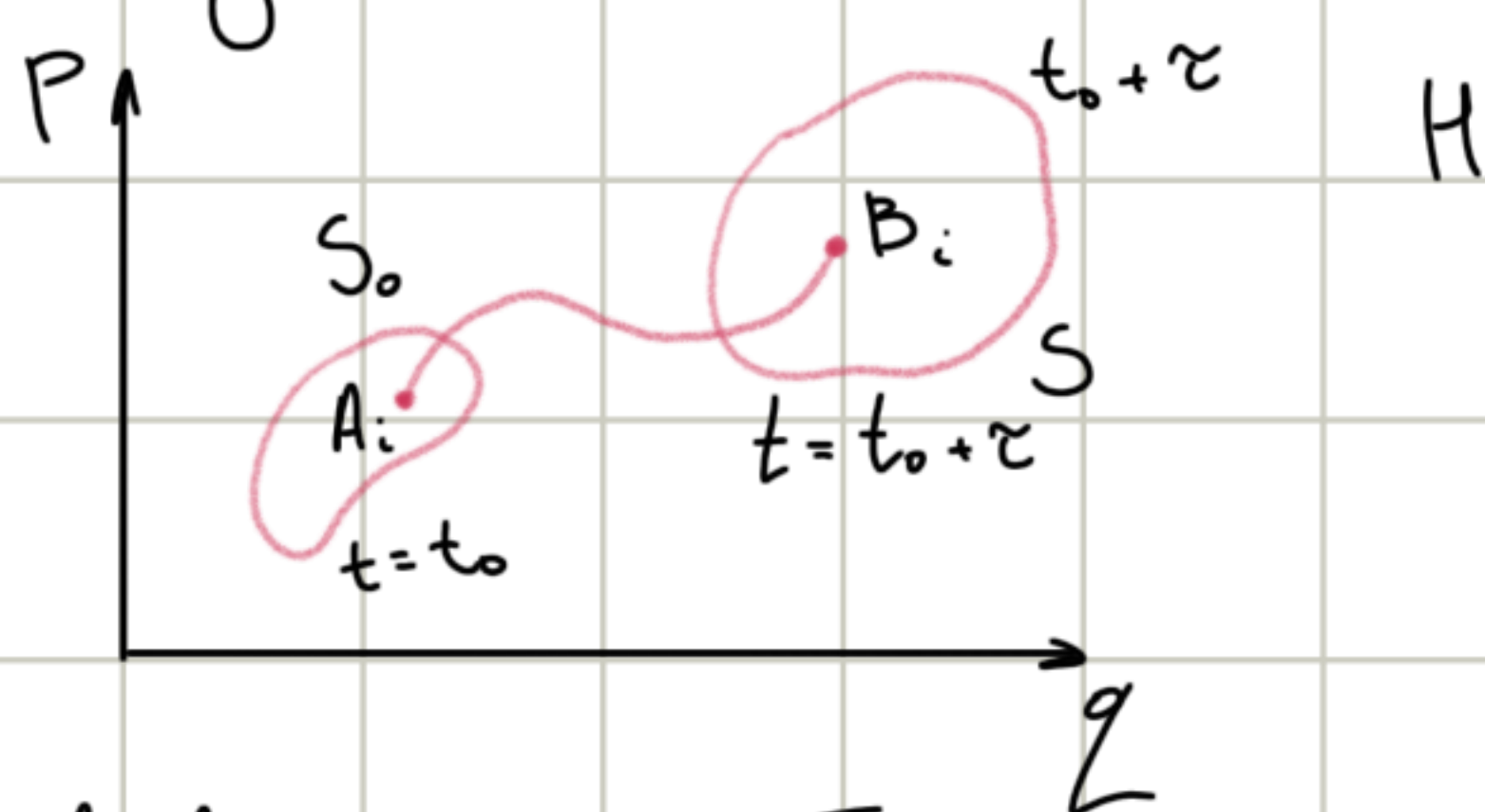


th лемма о сохранении фазового объема для рав. мсм.



$$J_0 = \int_{S_0} \dots \int dq_1 \dots dp_n$$

$$J(\tau) = \int_S \dots \int dq_1 \dots dp_n$$

th лемма о сoпp. ф. oбъема для рав. мсм.

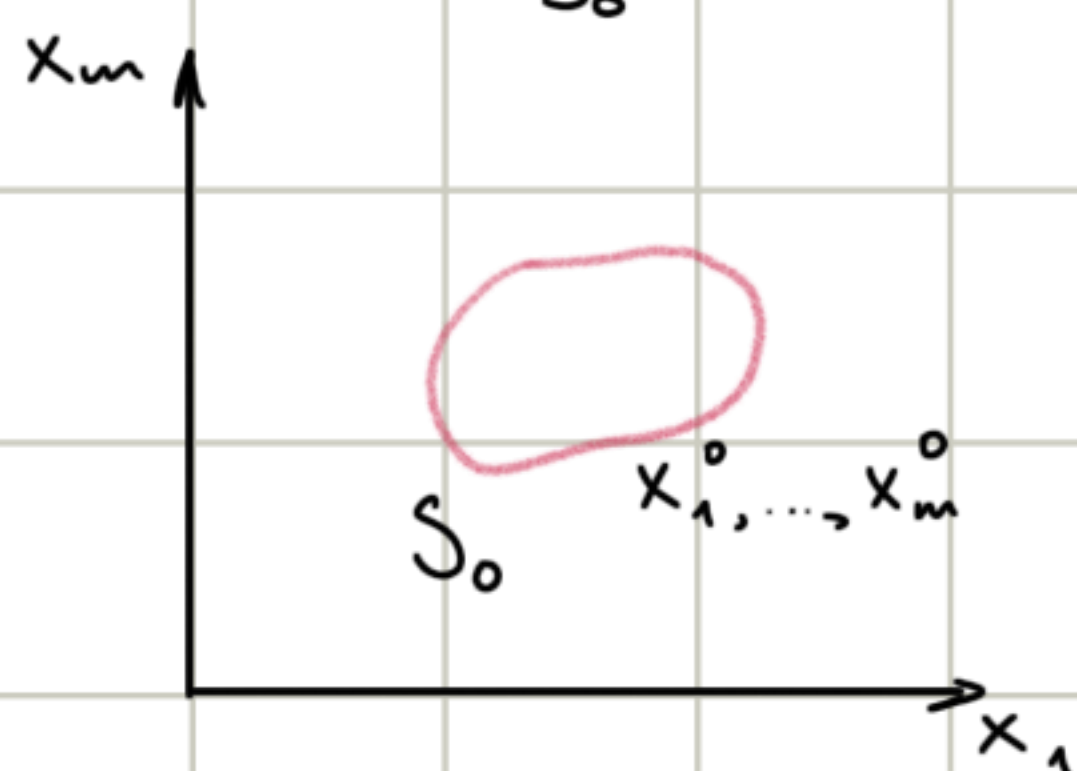
Универсальность $J(t)$ для рав. мсм. не зависит от времени, т.е. абс. инв. величина.

th лемма

В работе напомним ОДУ $\dot{\bar{x}} = \bar{f}(\bar{x}, t)$, $\bar{x} = \{x_1, \dots, x_m\}$ упрощ. ур.

$\text{div } \bar{f} = 0$ ($\text{div } \bar{f} = \sum_{i=1}^m \frac{\partial f_i}{\partial x_i}$), тогда фаз. объем сохр. на реш-ях.

$$\Delta. J_0 = \int_{S_0} \dots \int dx_1^0 \dots dx_m^0$$



$$J(t) = \int_S \dots \int dx_1^t \dots dx_m^t$$

$$x_i^t = \varphi_i(t, t_0, x_1^0, \dots, x_m^0)$$

$$J(t) = \int_{S_0} \dots \int |J| dx_1^0 \dots dx_m^0$$

$$|J| = \det \begin{pmatrix} \frac{\partial \varphi_1}{\partial x_1^0} & \dots & \frac{\partial \varphi_1}{\partial x_m^0} \\ \vdots & & \vdots \\ \frac{\partial \varphi_m}{\partial x_1^0} & \dots & \frac{\partial \varphi_m}{\partial x_m^0} \end{pmatrix}$$

$$|J|_{t=t_0} = |E| = 1$$

$$\frac{d|J|}{dt} = 0 \quad (\text{по К-м})$$

$$\frac{d|J|}{dt} = \sum_{i=1}^m J_i$$

$$J_i = \det \begin{pmatrix} \frac{\partial \varphi_1}{\partial x_1^0} & \dots & \frac{\partial \varphi_1}{\partial x_m^0} \\ \frac{d}{dt} \frac{\partial \varphi_i}{\partial x_1^0} & \dots & \frac{d}{dt} \frac{\partial \varphi_i}{\partial x_m^0} \\ \frac{\partial \varphi_m}{\partial x_1^0} & \dots & \frac{\partial \varphi_m}{\partial x_m^0} \end{pmatrix}$$

i-я строка

$$\frac{d}{dt} \frac{\partial \varphi_i}{\partial x_e^0} = \frac{\partial \varphi_i}{\partial x_e^0} = \frac{\partial f_i}{\partial x_e^0} = \sum_{k=1}^m \frac{\partial f_i}{\partial x_k} \frac{\partial x_k}{\partial x_e^0} = \sum_{k=1}^m \frac{\partial f_i}{\partial x_k} \frac{\partial \varphi_k}{\partial x_e^0}$$

$$J_i = \det \begin{pmatrix} \frac{\partial \varphi_1}{\partial x_1^0} & \dots & \frac{\partial \varphi_1}{\partial x_m^0} \\ \sum_{k=1}^m \frac{\partial f_i}{\partial x_k} \frac{\partial \varphi_k}{\partial x_1^0} & \dots & \sum_{k=1}^m \frac{\partial f_i}{\partial x_k} \frac{\partial \varphi_k}{\partial x_m^0} \\ \frac{\partial \varphi_m}{\partial x_1^0} & \dots & \frac{\partial \varphi_m}{\partial x_m^0} \end{pmatrix}$$

Умножим l-ю строку на $\frac{\partial f_i}{\partial x_e}$ → вычтем из i-й строки

$$J_i = \det \begin{pmatrix} \frac{\partial \varphi_1}{\partial x_1^0} & \dots & \frac{\partial \varphi_1}{\partial x_m^0} \\ \frac{\partial f_i}{\partial x_i} \frac{\partial \varphi_i}{\partial x_1^0} & \dots & \frac{\partial f_i}{\partial x_i} \frac{\partial \varphi_i}{\partial x_m^0} \\ \frac{\partial \varphi_m}{\partial x_1^0} & \dots & \frac{\partial \varphi_m}{\partial x_m^0} \end{pmatrix} = \frac{\partial f_i}{\partial x_i} |J| \Rightarrow \frac{d|J|}{dt} = |J| \sum_{i=1}^m \frac{\partial f_i}{\partial x_i} = |J| \cdot \text{div } \bar{f} = 0$$

$$|J| = \text{const} = 1$$

$$J(0) = \int_{S_0} \dots \int dx_1^0 \dots dx_m^0$$

$$J(t) = \int_S |dx_1 \dots dx_m| = \int_{S_0} |J| |dx_1^0 \dots dx_m^0|$$

$$\Rightarrow J(0) = J(t) \Rightarrow \text{фаз. объем сохр.} \blacktriangle$$

Для гами. сист.:

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad i = \overline{1, n}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}, \quad i = \overline{1, n}$$

$$\sum_{i=1}^n \frac{\partial}{\partial q_i} \frac{\partial H}{\partial p_i} + \sum_{i=1}^n \frac{\partial}{\partial p_i} \left(-\frac{\partial H}{\partial q_i} \right) = 0 \Rightarrow \text{для гами. сист. ф. } V \text{ сохр.} \blacktriangle$$

$$-\oint_C p \delta q = \iint_V \delta p \delta q$$

C-ма 1 степень свободы

V сохр. $\Leftrightarrow$ гами. сист.

Канонические преобразования

$$(*) \begin{cases} \tilde{q} = \tilde{q}(q, p, t) \\ \tilde{p} = \tilde{p}(q, p, t) \end{cases}$$

1) обратимое

$$q = q(\tilde{q}, \tilde{p}, t)$$

$$p = p(\tilde{q}, \tilde{p}, t)$$

2) $\nexists$ гами. сист. в гами. сист.

- каноническое преобразование

прим.

$$1) \tilde{q}_i = q_i, \quad \tilde{p}_i = p_i$$

$$2) \tilde{q}_i = \alpha q_i, \quad \tilde{p}_i = \beta p_i$$

$$\begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} \\ \dot{p}_i = -\frac{\partial H}{\partial q_i} \end{cases}$$

$$\begin{cases} \dot{\tilde{q}}_i = \alpha \dot{q}_i = \alpha \frac{\partial H}{\partial p_i} = \frac{\partial \tilde{H}}{\partial \tilde{p}_i} \\ \dot{\tilde{p}}_i = \beta \dot{p}_i = -\beta \frac{\partial H}{\partial q_i} = -\frac{\partial \tilde{H}}{\partial \tilde{q}_i} \end{cases}$$

$$\tilde{H} = \alpha \beta H\left(\frac{\tilde{q}}{\alpha}, \frac{\tilde{p}}{\beta}, t\right)$$

$$\frac{\partial \tilde{H}}{\partial \tilde{p}_i} = \alpha \beta \frac{\partial H}{\partial p_i} \frac{\partial p_i}{\partial \tilde{p}_i} = \alpha \frac{\partial H}{\partial p_i}$$

$$3) \tilde{q}_i = \alpha_i q_i, \quad \tilde{p}_i = \beta_i p_i \quad - \underline{\text{не кан.}}$$

$$4) \tilde{q}_i = p_i, \quad \tilde{p}_i = q_i \quad \text{к.н.}$$

критерий к.н.

Преобраз. $(*)$ гами. канонич. $\Leftrightarrow \exists F(q, p, t), \exists c:$

$$\sum_{i=1}^n \tilde{p}_i \delta \tilde{q}_i = c \sum_{i=1}^n p_i \delta q_i = -\delta F$$

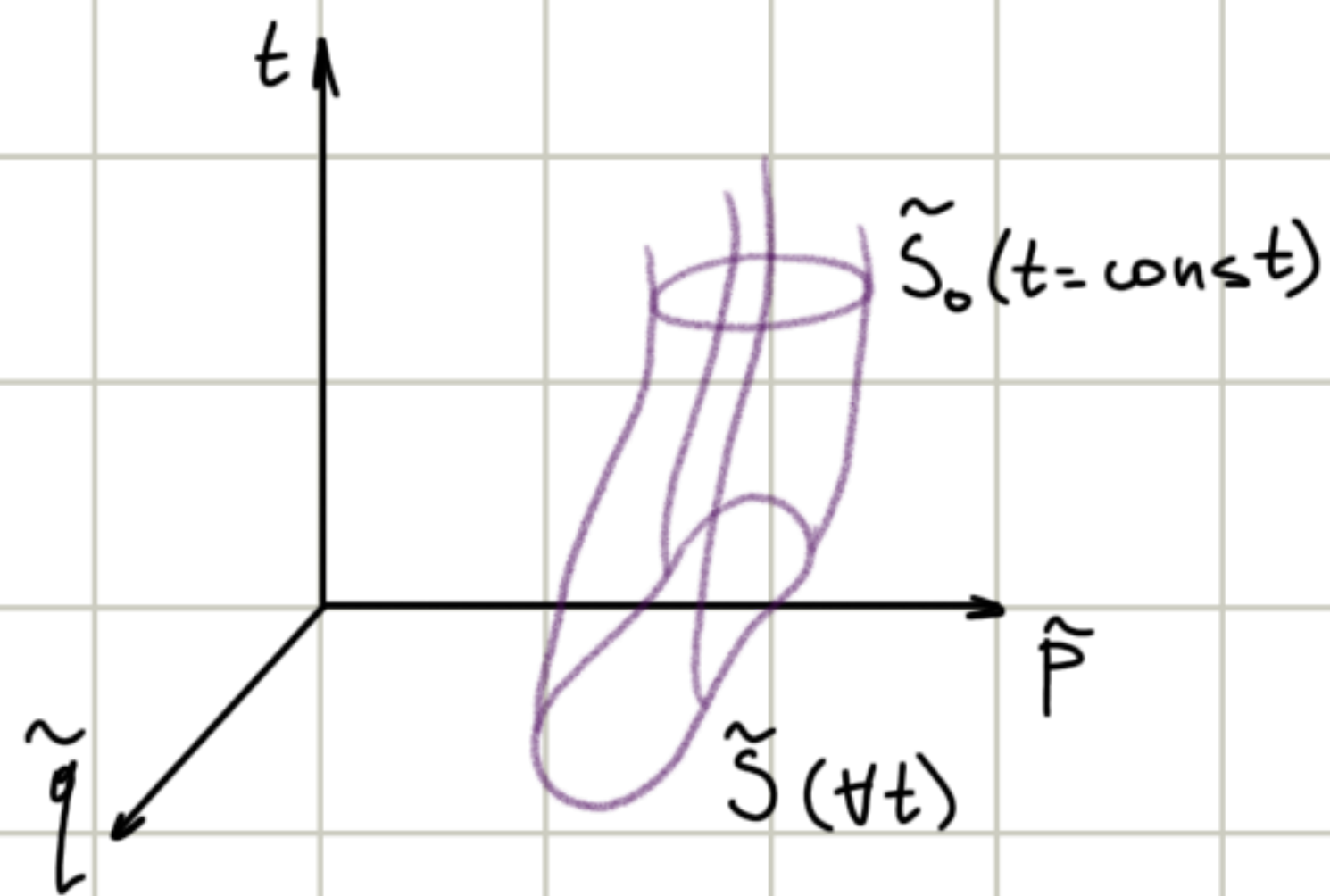
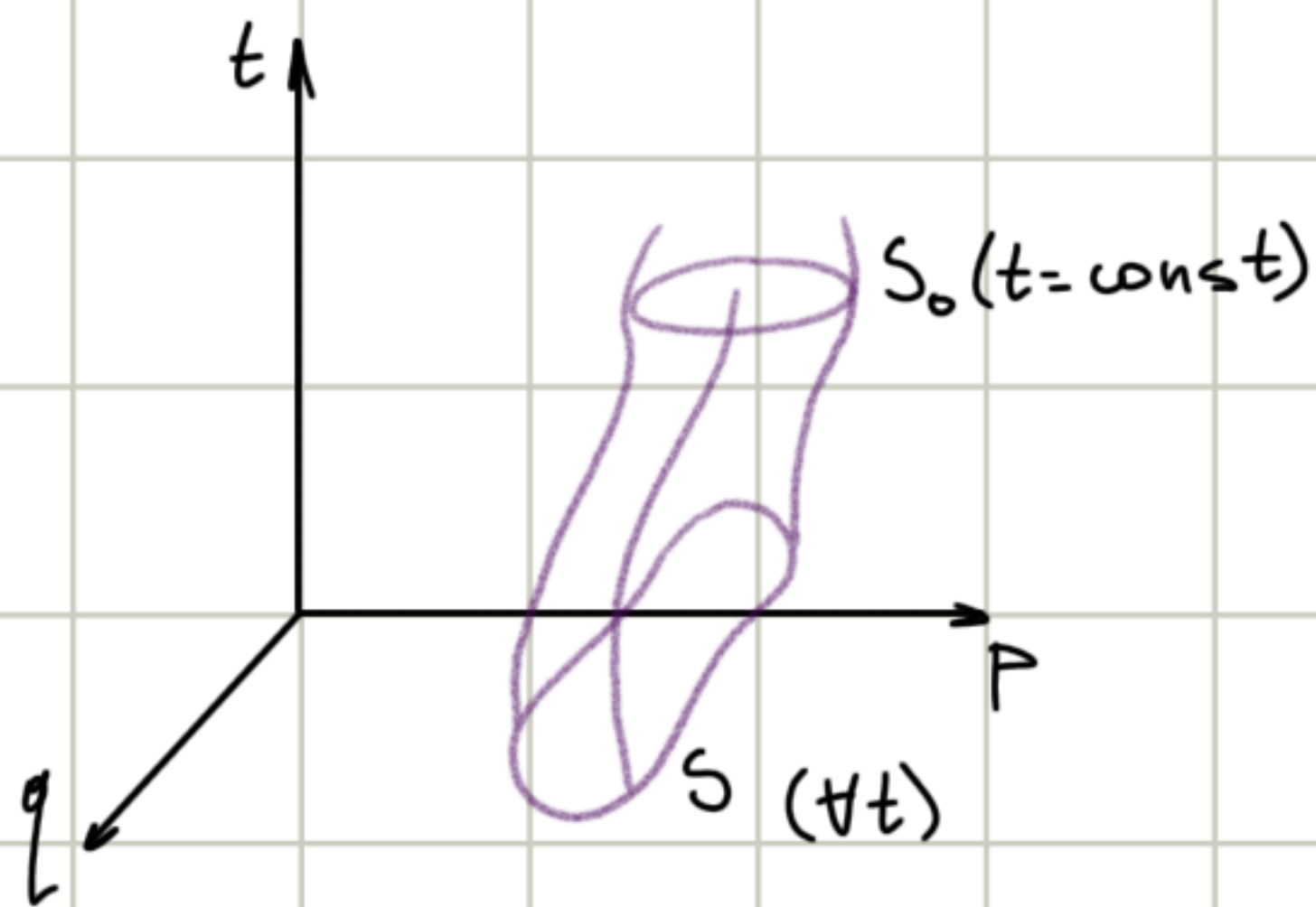
$$\delta q_i = dq_i \quad (\text{quasi var.})$$

$$\delta \tilde{q}_i = d\tilde{q}_i - \frac{\partial \tilde{q}_i}{\partial t} dt$$

$$\delta F = dF - \frac{\partial F}{\partial t} dt$$

Δ. 1) Каноничность

∃ преобр. кан. (∃ H, ∃ H̃)



$$\oint_{S_0} \sum_{i=1}^n p_i \delta q_i = \oint_S \left(\sum_{i=1}^n p_i dq_i - H dt \right)$$

$$\oint_{\tilde{S}_0} \sum_{i=1}^n \tilde{p}_i \delta \tilde{q}_i = \oint_{\tilde{S}} \left(\sum_{i=1}^n \tilde{p}_i d\tilde{q}_i - \tilde{H} dt \right)$$

$$\oint_{\tilde{S}_0} \sum \tilde{p}_i \delta \tilde{q}_i = \oint_{S_0} \sum \tilde{p}_i(\cdot) \delta \tilde{q}_i(\cdot) = \oint_{S_0} \sum \tilde{p}_i(\cdot) \sum_{i=1}^n \left(\frac{\partial \tilde{q}_i}{\partial q_i} \delta q_i + \frac{\partial \tilde{q}_i}{\partial p_i} \delta p_i \right) = c \oint_{S_0} \sum_{i=1}^n p_i \delta q_i$$

(th du Xyauuayua)

$$\oint_{\tilde{S}} \left(\sum \tilde{p}_i d\tilde{q}_i - \tilde{H} dt \right) = c \oint_S \left(\sum p_i dq_i - H dt \right)$$

$$\oint_S \left(\sum_{i=1}^n \tilde{p}_i(\cdot) d\tilde{q}_i(\cdot) - \tilde{H} dt \right) = c \oint_S \left(\sum p_i dq_i - H dt \right)$$

$$\oint_S \left[\sum_{i=1}^n \tilde{p}_i(\cdot) d\tilde{q}_i(\cdot) - \tilde{H} dt - c \left(\sum p_i dq_i - H dt \right) \right] = 0$$

$$\exists F: -dF = \sum_{i=1}^n \tilde{p}_i(\cdot) d\tilde{q}_i(\cdot) - \tilde{H} dt - c \left(\sum_{i=1}^n p_i dq_i - H dt \right)$$

$$(**) \sum \tilde{p}_i(\cdot) d\tilde{q}_i(\cdot) - \tilde{H} dt - c \left(\sum p_i dq_i - H dt \right) = -dF$$

∃ t - fix

$$d\tilde{q}_i \rightarrow \delta \tilde{q}_i \quad dt=0 \quad dF \rightarrow \delta F$$

$$\sum_{i=1}^n \tilde{p}_i \delta \tilde{q}_i - c \sum_{i=1}^n p_i \delta q_i = -\delta F$$

2) Каноничность

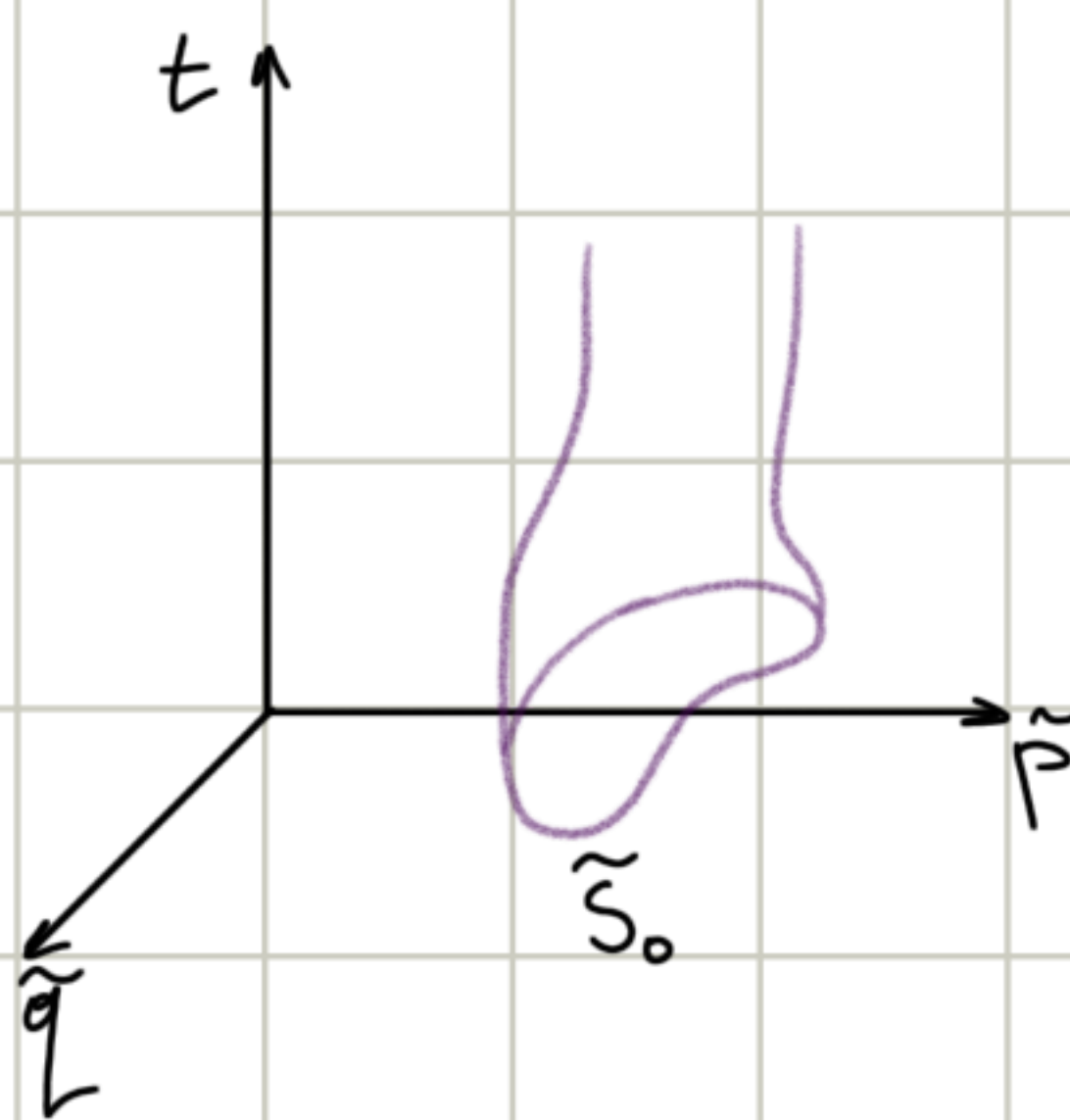
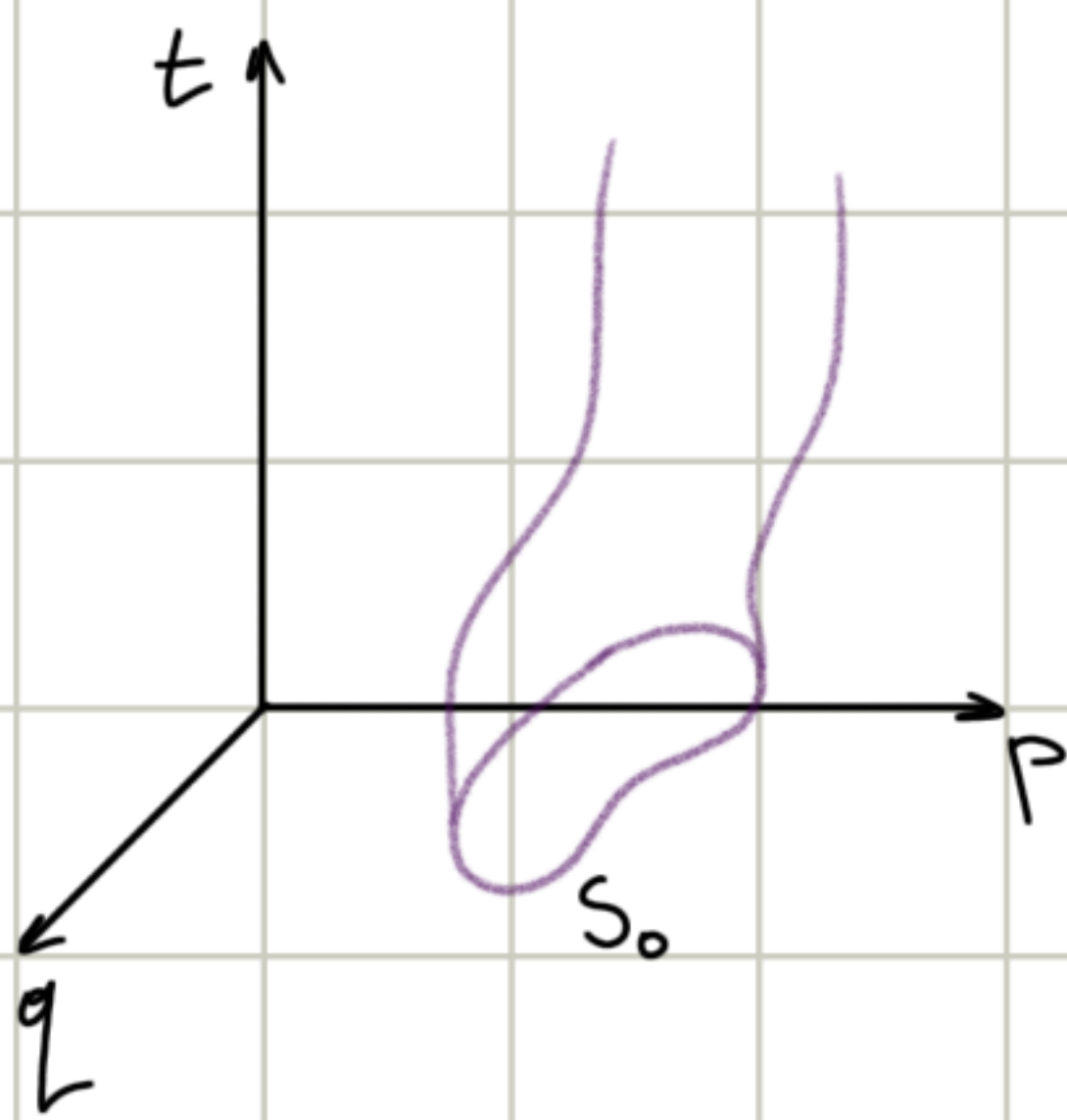
∃ канонично (**)

∃ F, c, H̃ (какая-то)

$$\tilde{q}_i = f_i(\tilde{q}, \tilde{p}, t)$$

$$\tilde{p}_i = g_i(\tilde{q}, \tilde{p}, t)$$

исм. группов
ное замены



$$\oint_{S_0} \sum \tilde{p}_i(\cdot) d\tilde{q}_i(\cdot) - \tilde{H} dt - c \left(\sum p_i dq_i - H dt \right) = - \oint_{S_0} dF = 0$$

$$\oint_{S_0} \left(\sum \tilde{p}_i(\cdot) d\tilde{q}_i(\cdot) - \tilde{H} dt \right) = c \oint_{S_0} \underbrace{\left(\sum p_i dq_i - H dt \right)}_{\text{инв. } p, q \text{ - зам}} \rightarrow \text{инв. } q \text{ на } S_0$$

$$\oint_{S_0} \left(\sum \tilde{p}_i(\cdot) d\tilde{q}_i(\cdot) - \tilde{H} dt \right) = \text{инв. } q \text{ на } S$$

$$\oint_{\tilde{S}_0} \left(\sum \tilde{p}_i d\tilde{q}_i - \tilde{H} dt \right) - \text{много инв.}$$

инв. $\Rightarrow$ зам. инв.

Рис.

1) F - произвольная ф-я

c - величина

Если $c=1$ - преобр. унитарное

2) $c \neq 0$

$\int c = 0$:

$$\sum \tilde{p}_i \delta \tilde{q}_i = - \delta F(\tilde{q}, \tilde{p}, t) = \sum_{i=1}^n - \frac{\partial F}{\partial \tilde{q}_i} \delta \tilde{q}_i - \frac{\partial F}{\partial \tilde{p}_i} \delta \tilde{p}_i$$

$$\begin{cases} \tilde{p}_i = - \frac{\partial F}{\partial \tilde{q}_i} \\ 0 = \frac{\partial F}{\partial \tilde{p}_i} \end{cases} \quad \begin{matrix} \frac{\partial^2 F}{\partial \tilde{p}_i \partial \tilde{q}_i} = -1 \\ \frac{\partial^2 F}{\partial \tilde{q}_i \partial \tilde{p}_i} = 0 \end{matrix} \quad \nexists F$$

3) Обрат. преобр. - кан., кан. $\frac{1}{c}$, произв. ф-я $-\frac{F}{c}$

$$\sum p_i \delta q_i = \frac{1}{c} \sum \tilde{p}_i \delta \tilde{q}_i = \frac{\delta F}{c}$$